

AUTONOMOUS AERIAL VEHICLE CHALLENGE 2026

International Academy of Aviation Industry (IAAI), KMITL Ladkrabang

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Technical Design & Analysis Report

Autonomous PX4 Hexacopter for Precision Fragile-Cargo Delivery

EFT X6100 hexacopter · Pixhawk 6X · Raspberry Pi CM4

Team AeroOptix

King Mongkut's University of Technology North Bangkok

Faculty of Engineering

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Rules & Regulations V1.3 (July 2026), as amended by the

24 July and 28 August 2026 event briefings

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Document version 2.0 · revised 28 August 2026, after the 09:00 competition-day briefing.

Version 1.1 (17 July 2026) described a 500 mm **quadcopter** flying **one egg per sortie** under a 20 m ceiling, and had never flown outdoors. Every section below has been rebuilt against the aircraft that is actually flying, the envelope the committee actually briefed, and the flights that have actually been flown. Where something is still unproven this report says so in the same sentence as the claim.

Structured to rules Appendix A (Executive Summary 10 pts · Team Organization 10 pts · Concept Development 30 pts · Engineering Design & Analysis 40 pts), and cross-referenced to the Appendix B DCOS airworthiness criteria (§3.8).

0. What changed since version 1.1

Area	v1.1 (17 Jul 2026)	This document (28 Aug 2026)	Why
Airframe	Holybro X500 V2 quad-X, 500 mm, ~2.6 kg	EFT X6100 hexacopter , 1,000 m wheelbase, 18-inch props, 7.2 kg AUV	The quad was never procured; DRON (Defence Research Operation Network) lent the X6100 frame, ESCs and props on 22 Jul 2026
Mission shape	1 egg per sortie, up to 4 sorties	4 eggs in ONE flight ; FLIGHT $\supset$ DELIVERY, scored per delivery	24 Jul 2026 event-briefing override
Ceiling	20 m AGL	30 m AGL ; search band 10–30 m; transit unchanged at 20 m	28 Aug 2026 competition-day briefing
Transit corridor	PDF P1→P2→P3	P1→P2'→P3' , moved 14–23 m west of the main building	Briefing: the PDF's north leg ended on the building roof
Search area	Full rules polygon	Rules polygon cut at E 110 m (west of the building)	Briefing declared three no-fly bands, one of them mid-polygon
Failsafe RTL altitude	20 m	25 m	Briefing: “25 m clears everything on the field”
Sweep	12 m, decode in flight	20 m white-pad finder + id read on 10.5 m decode visits	Obstacle clearance over the display aircraft; 20 m puts the marker under the decode floor, the pad well above the blob floor
Marker set	ids 1–6	ids 0–6 (seven)	The rules' Figure 7 encodes 1, 2, 0, 4, 5, 6 — id 0 is a real pad
Payload	One servo on AUX 9	Four latches on AUX 4/1/2/3 , DO_SET_ACTUATOR, diagonal release order	Four eggs per flight; PX4 has no DO_SET_SERVO handler
Camera	1280 px, “~99.7°” placeholder, on a gimbal	1280 px, 74.2° measured , hard-mounted , mount yaw 180° measured	Both numbers were assumptions; both were wrong; both are now bench measurements
Height reference	Optical flow + TF-Luna	Barometer (EKF2_HGT_REF=0) + Benewake TFmini-S ; no optical flow	No flow module in the kit; GPS height reference diverged 10.8 m peak-to-peak in one flight
Battery	6S 7500 mAh LiPo	6S 17,000 mAh semi-solid (+ a 15,000 mAh spare)	7500 could not finish the mission
Validation	SITL only	SITL plus 10 outdoor flights at the KMUTNB practice field (20, 26, 27 Aug)	The system has now flown a complete mission end to end
Test suite	239 tests	771 tests	

Executive Summary (10 pts)

We field a single **autonomous PX4 hexacopter** that carries **all four committee-assigned eggs in one flight**, lands on each pad whose ArUco marker id was assigned to us, and releases that egg **only after touchdown is confirmed** — inside the 20-minute operation window at the IAAI KMITL field.

The design philosophy is **fast-but-safe determinism**. A classical computer-vision pipeline — no machine learning anywhere in the flight loop, no network of any kind (network use is an automatic disqualification) — runs on a Raspberry

Pi CM4 companion computer and commands a Pixhawk 6X flight controller. The aircraft takes off from the Launch & Recovery point, flies the mandatory transit corridor at a strict 20 m, surveys the search area once, descends over each assigned pad in verified altitude rungs, lands on it, releases the egg on the ground, and flies the corridor home to land and disarm.

Cargo location is unknown at take-off. The committee assigns marker ids, not coordinates. The system therefore performs a **blind visual search with a pad registry**: one boustrophedon sweep decodes every pad it can see into an id-keyed registry, undecoded white-pad candidates are revisited at the legal 10 m floor to read their ids, and the flight then serves its four assigned pads directly. The sweep stops early as soon as every id this flight owes is confirmed.

Envelope as amended at the 28 August 09:00 briefing: ceiling **30 m** AGL, transit strictly **20 m**, search band **10–30 m**, below 10 m only for the delivery descent over a pad, failsafe RTL at **25 m**, and a transit corridor moved west of the main building. The flown search area is the rules' polygon cut at E 110 m so that every straight-line move the mission can make — pad hops, decode visits, egress, even a failsafe RTL — stays clear of the three no-fly bands the committee declared.

Validated results.

Evidence	Result
Real flight, complete mission (KMUTNB practice field, 27 Aug 2026, 15:25, 4 min 12 s)	Transit ingress 3/3 → sweep found 5 pads in ~20 s → early-stop on the assigned ids → land-ON + release on pad 1 (0.14 m from centre) and pad 4 → egress → landed 0.7 m from the launch point. 2/2 eggs delivered, both after touchdown.
Real flight, full survey (27 Aug 2026, 12:44, 5 min 56 s)	7-leg sweep, 6,044 frames, 5 of 6 pads decoded and confirmed ; the sixth was printed with marker id 0 , which the id filter then still excluded — the reason the valid set is now 0–6
SITL, competition envelope (24 Aug 2026, transit 20 m / ceiling 20 / sweep 12 m — the hardest decode case)	6/6 pads found, 6/6 ids correct, 4/4 eggs delivered , releases 0.09–0.22 m from truth, transit 6/6, landed 2.0 m from L&R, 536 s of the 1200 s window
SITL, four eggs in one flight (25 Jul 2026)	4/4 delivered, releases 0.15–0.21 m, max altitude 19.69 m, 457 s , verifier 14 checks / 0 warnings
Software integrity	771 automated tests green; lint + type-check clean; the flight core is deterministic and offline

The honest status. Three things are not yet proven and are stated here rather than buried: (1) the 28 August briefing changes — new corridor, 30 m ceiling, 20 m sweep, cut search area — are **unflown**, and the 13:00–18:00 trial slot is their only test before scoring; (2) **battery endurance is the binding risk**: the gauge falls 7–8 percentage points per minute under load, so a full pack reaches the 15 % return floor at roughly minute 8.5–10 against a plan that needs 10–13 minutes of flying — the mission is designed to give up eggs safely rather than the aircraft, and a second charged pack plus an automatic recovery flight is the answer; (3) the committee has not published coordinates for the new corridor or the no-fly bands, so ours are georeferenced from briefing photographs to **±5 m** and will be replaced by the committee's if they are issued.

1. Team Organization (10 pts)

Team **AeroOptix**, Faculty of Engineering, KMUTNB — a **first-time AAVC entrant**. Four months ago the team had no aircraft, no budget, and no prior autonomous-drone build. The system described here was built from zero in that time, on equipment donated, lent, or paid for by the members themselves.

Role	Function
Team Leader (contact)	Overall coordination; single point of contact with AAVC staff
Faculty Advisor	Airworthiness sign-off, safety oversight
Mission software	Orchestrator, search/serve logic, time and energy policy, audit trail
Flight systems / Safety Pilot	Pixhawk 6X integration, PX4 parameters and failsafes, manual takeover and kill authority
Vision & GCS	ArUco pipeline, camera calibration and exposure, ground station
Airframe & power	Frame build, payload rack, wiring, battery and mass budget
C2 / GCS operators (≤4)	Per-flight GO, monitoring, resupply coordination

Crew inside the GCS operating area complies with the rules limit: **1 safety pilot, ≤4 command-and-control operators, ≤3 technical-support personnel**. All flight-line crew wear PPE (protective glasses, reflective vests) per DCOS. The aircraft frame, ESCs, propellers and both battery packs were provided by **DRON (Defence Research Operation Network)**, whose support is acknowledged.

2. Concept Development (30 pts)

2.1 Capability parameters derived from the operational challenge

The operating environment fixes the design envelope. From the rules (V1.3), the two event briefings, and the IAAI KMITL field geometry:

Environment / rule driver	Value	Derived vehicle requirement
Controlled airspace (geofence)	~296 × 172 m	Endurance and speed to cross it repeatedly inside the window
Search area as flown	Rules polygon cut at E 110 m — the open field west of the main building, ~69 × 60 m	Cover ~0.4 ha with a decodable sensor footprint
No-fly bands (briefing)	Main building E 122–186 / N 25–84; east end E 215–256 / N 37–116; orange block E 219–255 / N –14...40 (ENU metres from L&R, ±5 m)	Every straight-line move must stay west of E 110 m
Transit altitude (mandatory, both directions, scored per point)	strictly 20 m	Precise altitude hold at 20 m AGL in an absolute frame that does not drift
Ceiling (briefing)	30 m AGL	Companion watchdog warns at 30.5 m, returns home above 31.5 m sustained
Search band	10–30 m AGL	Pad acquisition from 20 m, id decode from 10.5 m
Delivery descent	below 10 m only, over the pad	Precision terminal descent ending in a land-ON
Landing pad	1 × 1 m white square, 750 mm dia. black circle, 400 mm ArUco (DICT_4X4_50)	Decode a 400 mm marker from the legal band with a 1280 px sensor
Marker set	ids 0–6 — six pads placed, four assigned per team	Reject the two distractor ids for the whole mission
Operation window	5 min setup + 20 min operation , per-minute overtime penalty	A full four-egg flight must fit with reserve
Cargo	Four organiser boxes (heart, ~16 × 7 × 18 cm, 300-gsm), one no. 0 raw egg each	Four independent holds; gentle touchdown; release on the ground
MTOW limit	≤ 25 kg	7.2 kg leaves large margin
No RTK GPS	~0.5–1.5 m, plus a common-mode bias against the published coordinates	Vision owns the final metre; GPS never commits the egg

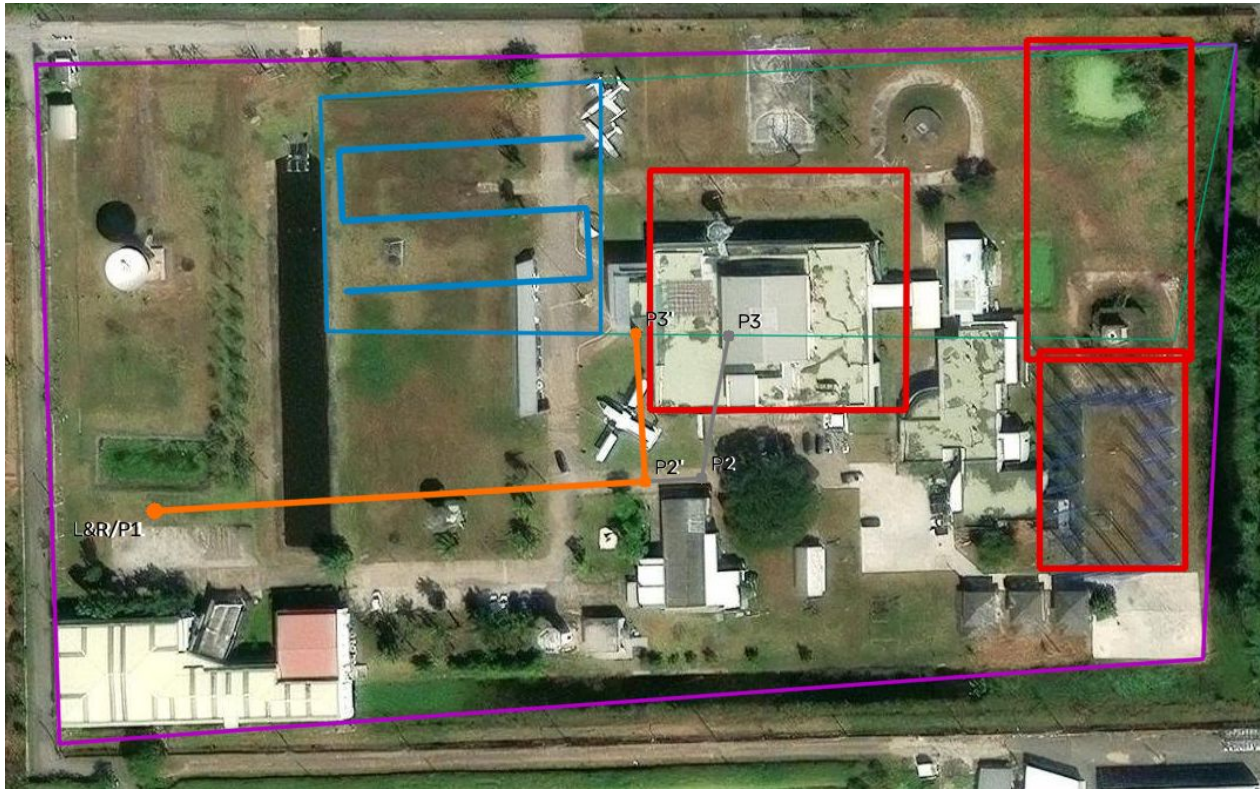


Figure 1. The competition field as briefed on 28 August 2026, georeferenced against the ground station's cached satellite imagery. **Purple** — the controlled airspace (geofence). **Teal** — the rules' search polygon. **Blue** — the three sweep legs actually flown, over the polygon cut at E 110 m. **Orange** — the briefed transit corridor L&R/P1 → P2' → P3'. **Grey** — the corridor the PDF specifies (P2, P3), whose north leg ends on the main building's roof. **Red** — the three no-fly bands declared at the briefing: the main building, the east end of the search area, and the rules' orange block. Every straight-line move the mission can make stays west of the building box.

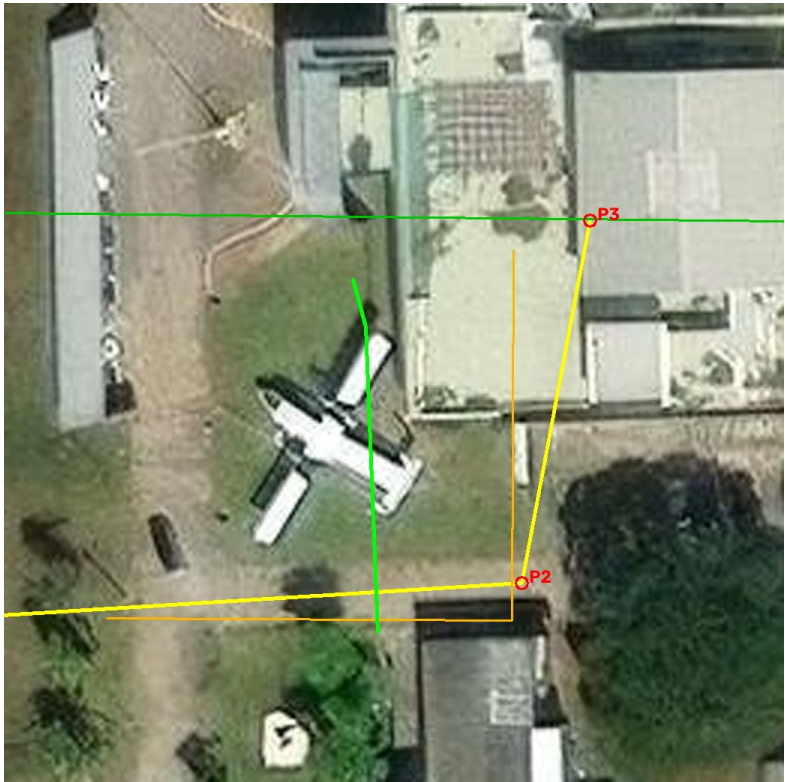


Figure 2. The corridor region at 4x. **Yellow** — the PDF’s P1→P2→P3 with P2 and P3 marked; the leg between them crosses the building. **Orange** — the route drawn on the committee’s own briefing slide. **Green** — the operator’s traced corridor, running up the grass strip between the parked display aircraft and the building’s west wall; P2’ and P3’ are taken from this line and are accurate to ±5 m. Committee-published coordinates, if issued, replace ours.

Decode geometry drives the sweep plan. With the **measured** 74.2° lens on a 1280 px-wide sensor, the ground swath is $2 \cdot h \cdot \tan(37.1^\circ)$ and marker pixels are $1280 \times 0.4 / \text{swath}$:

Altitude	Swath	400 mm marker	1 m pad	Role
20 m (sweep)	30.3 m	16.9 px — below the 18 px decode floor	42 px — well above the 18 px blob floor	White-pad finder: 3 legs, 216 m, 87 s over the cut polygon
10.5 m (decode visit)	15.9 m	32 px — inside the 25–35 px band proven in flight	80 px	Id reader: hover and dwell until the marker decodes (~47 s)
12 m (the alternative)	18.2 m	28.2 px	70 px	Would decode in flight, but 5 legs / 344 m / 142 s and lower clearance over the display aircraft

Flying the survey at 20 m and paying for a decode visit only where a white pad was actually seen is **55 s cheaper** than sweeping at 12 m on this polygon, and keeps the aircraft above the field obstacles the committee flagged.

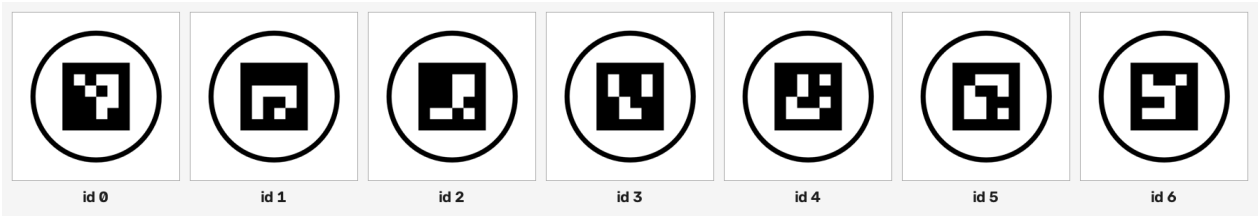


Figure 3. The seven landing-pad faces the mission must tell apart — 1 x 1 m white square, 750 mm diameter ring, 400 mm DICT_4X4_50 marker. Rendered here by `vision.detectors.aruco.render_pad_bgr`, the same function that generates the simulator’s pad textures and the detector’s test images, so what the detector is tested against is what the simulator shows. Six of these are placed on the field and four are assigned to us; **id 0 is a real pad** — the rules’ Figure 7 encodes 1, 2, 0, 4, 5, 6.

Window budget drives the mission shape. A full four-egg flight is budgeted at **894 s of the 1200 s window** (17 s take-off + 53 s ingress + 24 s climb/transition + 87 s sweep + 188 s decode visits + 420 s of four serves + 17 s + 53 s

egress + 35 s landing). The last delivery's gate therefore runs with **516 s remaining** against a 300 s floor. Carrying all four eggs in one flight — rather than four one-egg flights — is what buys that margin, and the briefing explicitly permits it.

2.2 Selected delivery system — configuration and architecture

Vehicle configuration: single multirotor, hexa-X. The mission is a confined urban field with a vertical take-off and landing requirement, a 1 m landing target, and a fragile payload; a multirotor is the only sensible class, and six rotors were chosen over four for three reasons that matter operationally:

1. **Rotor-out tolerance.** With one motor lost, PX4's sequential-desaturation allocator mixes roll, pitch and thrust before yaw, so a hexacopter keeps attitude and thrust authority and gives up only yaw. That is recoverable by the safety pilot; on a quad it is not. (Measured in SITL: unallocated yaw torque rises to 0.25 with the allocator still solving for the other axes.)
2. **Payload and mass margin.** Four boxed eggs plus a four-latch rack on a 1.0 m frame with 18-inch props hovers at ~31 % of available thrust, so the motors loaf and retain full authority against gusts.
3. **Availability.** The X6100 airframe, ESCs and props were lent complete, which converted a procurement risk into flight time.

A single aircraft — rather than a cooperative search-and-deliver pair — keeps mass, cost and failure surface small, and the registry-based search removes the need for a second vehicle to find pads.

System architecture.

AIR SEGMENT	GROUND SEGMENT
Pixhawk 6X (PX4 v1.17)	Safety-pilot RC (ELRS)
attitude/position control, geofence,	RC + manual takeover, kill
FC-level failsafes, RTL	
^ MAVLink (serial)	MAVLink Web GCS (Svelte + FastAPI)
	on ONE live map, camera, plan,
Raspberry Pi CM4 (companion)	radio 4-of-6 queue editor, GO,
mission orchestrator (async Python)	confirmed-pad readout, KILL
vision worker (OpenCV ArUco)	WiFi
safety watchdog (2 Hz)	
audit trail + frame recorder	(CM4 raises its own local AP
^	- no internet, no 4G)
OV9281 nadir camera (1280x720 mono,	
global shutter, hard-mounted)	FPV camera + 5.8 GHz VTX
TFmini-S lidar · non-RTK GNSS	-> the "transmit" half of the
4 egg latches (AUX 4/1/2/3)	imaging requirement

The flight-management stack is a **non-proprietary, open-source architecture** (PX4 on Pixhawk 6X) integrated by the team, which complies with the rules' prohibition on complete proprietary flight packages. All higher-level autonomy runs on the CM4 as deterministic classical code that the team wrote.

Rationale for the key decisions:

- *Classical CV, never ML, in the flight loop.* The target is a fiducial marker; an OpenCV ArUco decode is deterministic, cheap on a CM4, and auditable frame by frame. No torch, no YOLO, no VLM, no cloud, no network.
- *Land-ON, then release.* Scoring pays for the egg being placed on the pad intact. Releasing from the air risks a broken egg for zero payload score, so the release is hard-gated behind a confirmed touchdown and is skipped and audited if telemetry says the aircraft is airborne.
- *The id is the authority, not the position.* A pad that decodes to the wrong id can never steer a descent, and the LAND gate refuses to commit unless the assigned id was decoded during the approach.
- *Vision owns the final metre.* Without RTK the GNSS fix wanders by more than the pad is wide, so the terminal descent is closed by the camera.
- *The GCS monitors; it does not fly.* One operator click per flight starts it; everything after that is on board, so a WiFi dropout costs a picture, never the mission.

2.3 Concept of operations

A **FLIGHT** is one arm→disarm cycle. A **DELIVERY** is one pad served inside a flight, and each delivery is scored independently. With four eggs aboard, the whole mission is normally **one flight containing four deliveries**:

```

PREFLIGHT  operator loads the 4 assigned ids as an ordered queue, confirms the
            eggs are racked and the crew is clear -> ONE "GO" click
-> TAKEOFF  arm, climb out, then close the last metres en route
-> TRANSIT_INGRESS  P1 -> P2' -> P3' at 20 m (scored per coordinate)
-> SEARCH   boustrophedon sweep of the cut search polygon at 20 m,
            registering every white pad seen; decode visits at 10.5 m
            read the ids; the sweep stops as soon as every id this
            flight owes is confirmed

```



```

-> for each assigned id, in turn (one DELIVERY each):
    LOCALIZE    hop to the pad at sweep altitude, descend in verified
                  altitude rungs, re-centring on the decoded marker
    LAND        land ON the pad (final rung tolerance 0.2 m on a 1 m pad)
    DROP        after touchdown confirms + 2 s settle, open that egg's latch
-> TRANSIT_EGRESS P3' -> P2' -> P1 at 20 m (scored per coordinate)
-> LAND at L&R -> DISARM (resupply crew may approach)

```

Search-and-identify tactic. A flight whose assigned ids are not all in the registry flies the sweep. Every pad the camera sees is clustered by decoded id with a k-vote confirmation, so a single mis-decode cannot create a target. A pad seen as a white blob but not decoded is revisited at 10.5 m and the aircraft dwells until the marker reads; a pad decoded but short of its vote quota gets a cheap vote top-up visit instead of a re-sweep. The registry persists across flights, so a later flight whose whole assignment is already registered flies **direct**, with no sweep at all.

Delivery tactic (land-ON, touchdown-gated). Over the assigned pad the aircraft descends through altitude rungs, re-centring on the decoded marker at each rung with a tightening tolerance down to **0.2 m** before the final descent. Three gates protect the egg:

- the **id-verified LAND gate** refuses to land unless the delivery's own assigned id was decoded during the approach — a wrong-id pad or an undecoded blob can never commit the egg;
- the **centred-final-rung gate** refuses to land if the re-centre never converges, climbing back and deferring instead, with one re-approach if the window allows;
- the **touchdown gate** releases only after PX4 reports the vehicle landed, plus a 2 s settle; if telemetry reads airborne at the release point the release is skipped and audited.

Between deliveries inside a flight the aircraft stays **armed** on the pad (`COM_DISARM_LAND = -1`), so no re-arm happens over the field and PX4's home point stays pinned to the launch site.

Repeat-delivery and resupply tactic. Between flights the aircraft lands at L&R and disarms so the crew can approach. The 20-minute window clock starts at the **first GO**. A per-flight gate refuses a launch that cannot finish inside the window unless the operator explicitly accepts the overtime penalty, and a **per-delivery gate** re-checks time and battery before **every** egg: failing it skips the remaining eggs of that flight and brings them home rather than starting a descent the budget cannot finish.

Planned battery egress — the mission gives up eggs, not the aircraft. Below **30 %** indicated the flight starts nothing new: the sweep stops, decode visits are skipped, the delivery gate refuses, and the aircraft flies the **normal egress through the corridor**, lands and disarms. The crew swaps to the second pack and the **recovery flight** serves what is still owed — flying direct off the registry, and firing only the latches that still hold eggs (the crew does not re-rack eggs at a battery swap, so the recovery flight continues the wiring order through the unfired holds). The 15 % return-to-home and 10 % land-in-place thresholds sit **underneath** this as failsafes; the point of the planned egress is never to reach them.

3. Engineering Design & Analysis (40 pts)

3.1 Airframe structure and integrity

- **Frame:** EFT X6100 hexa-X, **1.000 m wheelbase** (0.500 m arm radius), folding carbon arms and carbon plates, frame mass ~2.75 kg, manufacturer MTOW ~12 kg. Provided by DRON.
- **Rotor clearance:** adjacent motors sit 500 mm apart on the hexagon; with **18 × 6.5 in** propellers (457 mm diameter) the tip-to-tip gap is **~43 mm**, so there is no rotor overlap and no interaction between discs (DCOS “clearance between moving parts”).
- **Payload rack:** four independent holds arranged front-left, front-right, rear-left and rear-right about the centre of gravity, each sized to accept the organiser's heart-shaped box (~16 × 7 × 18 cm, 300-gsm) rather than a bare egg. The release order is **diagonal** so the net CG moment is zero when full and zero again after the first two eggs leave.
- **Ground clearance:** the camera and the lidar are carried 0.25 m below the CG on the underside; the landing gear keeps both, and the loaded boxes, clear of the pad at touchdown.
- **Structural load cases:** flight manoeuvre load is envelope-limited to **30° tilt** (`MPC_TILTMAX_AIR`) and **2 m/s² horizontal acceleration** (`MPC_ACC_HOR`); the land-ON touchdown is absorbed at **MPC_LAND_SPEED = 0.3 m/s** with the final autonomous descent pinned to **0.4 m/s** (`MPC_Z_V_AUTO_DN`); ground handling and transport use the folded arms.
- **Power path:** the motors are fed from a separate distribution board that the flight controller cannot sense. A Holybro **PM02D** supplies the Pixhawk 6X and avionics only, and the CM4 runs from its own power-bank rail so companion current cannot brown out the flight controller. This split is deliberate and has one important consequence, carried through §3.2: **any current the FC reports is avionics draw, never flight current**, so the

battery gauge must never be current-fused.

- **Mass, as weighed 19 August 2026: 7.2 kg all-up, ready to fly**, with one 17,000 mAh pack (1.5 kg) and four boxed eggs aboard — **29 % of the 25 kg rules limit**. Rule adopted after three conflicting weighings in one afternoon: *never record a mass without recording what was on the aircraft* (pack count, power module, companion, eggs).
- **Open items (honest status):** a formal **CG placement record** and a **quantified structural load-factor margin** are not documented; the aircraft is trimmed by flight behaviour, not by a calculated CG envelope. The organiser's boxes are heavier than the 50 g dummies in the weighed figure, so the aircraft is re-weighed with the real cargo at the field committee's scale.

3.2 Propulsion and flight-performance analysis

Parameter	Value	Basis
Airframe	EFT X6100 hexa-X, 1.000 m wheelbase, 18 × 6.5 in props	as built
Motors	6 × EFT E5 5008, 335 KV	as built
ESC	6 × PWM-only ESC (no telemetry lead — see §3.3)	as built
Max thrust per motor	37.65 N (3,838 gf at 100 % throttle, 24 V, 18 × 6.5 in)	manufacturer bench table (Power-System-Guide)
Max thrust, 6 rotors	226 N ≈ 23 kgf	6 × above
All-up weight (flight-ready, 4 boxed eggs)	7.2 kg	weighed 2026-08-19
Thrust-to-weight	≈ 3.2 : 1	23 kgf / 7.2 kg
Hover thrust fraction	≈ 31 % of maximum (11.8 N per motor)	as above
Hover current	≈ 30 A at 6S (~670 W)	E5 bench anchor scaled by mass ^{1.5}
Hover throttle, measured in flight	0.59–0.64	ULog, 2026-08-27
Battery (mission)	6S 17,000 mAh semi-solid , 10 C, 1.5 kg, 25.1 V full (4.18 V/cell), ~22.6 V empty (3.77 V/cell)	as fitted 2026-08-19
Battery (spare, for the swap)	6S 15,000 mAh	as held
Usable energy	12,750 mAh (75 % of nameplate)	policy
Planned cost of a four-delivery flight	4,700 mAh (1,750 first delivery + 3 × 900 + 250 margin)	battery: seeds, deliberately conservative
Cruise / sweep speed	3.0 m/s (MPC_XY_CRUISE), max 5.0 m/s (MPC_XY_VEL_MAX)	config
Climb / take-off	1.5 m/s (MPC_Z_VEL_MAX_UP, MPC_TKO_SPEED)	config
Autonomous descent / touchdown	0.4 m/s (MPC_Z_V_AUTO_DN) / 0.3 m/s (MPC_LAND_SPEED)	config
Auto yaw rate	25 °/s (MPC_YAWRAUTO_MAX)	config

Excess-thrust margin (DCOS). Hovering at ~31 % of available thrust leaves roughly a two-thirds throttle headroom, so the motors run cool, the allocator never saturates (confirmed in the 27 August logs), and full control authority remains for gust rejection. The margin holds across the loaded and empty cases because four boxed eggs are a small fraction of a 7.2 kg aircraft.

Energy — the binding constraint, stated plainly. On paper the budget is comfortable: 4,700 mAh planned against 12,750 mAh usable. In flight it is not that simple, because this aircraft **cannot measure current**. With BAT1_CAPACITY = -1 the state of charge is derived from voltage alone, and PX4 only load-compensates when it has a current reading — which it does not — so the indicated percentage sags under thrust and rebounds when the aircraft settles. Measured across four flights on 27 August:

Observation	Value
Continuous sweep flight, full pack	93 % → 26 % indicated in 5.8 min
Three flights with pauses, full pack	96 % → 25 % indicated over 8.6 min of flying
Indicated drain under load	~7–8 percentage points per minute
Extrapolated time to the 15 % return floor	minute 8.5–10 of powered flight

Observation	Value
KMITL plan	894 s budgeted , ~10–13 min realistic
Cell voltage at 25 % indicated, rested	3.87–3.92 V/cell = true state of charge ~55–65 %

The gauge is therefore **pessimistic** — the pack almost certainly holds the energy (roughly 10 Ah of 17 Ah for a 15-minute flight at ~40 A) — but the **true capacity has never been measured**, so the flight is planned against the gauge, not against the optimistic interpretation. Consequences, all already implemented: the per-delivery gate gives up eggs 3 and 4 safely rather than the aircraft; the planned 30 % egress flies the scored corridor home instead of falling into a straight-line failsafe RTL; and a **second charged pack with an automatic recovery flight** is the primary mitigation. The one measurement that closes the question is the **mAh returned by the charger** after a flight, which is recorded from now on. The battery percentage floors were **not** lowered further: the ESCs’ low-voltage cut-off is unknown, and a floor is worth more than an egg.

Flight-performance verification against §2.1. Requirements met: the survey covers the cut polygon in 3 legs at 3 m/s; transit holds 20 m in the barometric frame (commanded 19.5 m — see §3.3 — measured maximum 19.69 m in the four-egg SITL run, no ceiling flag); and a complete real mission — transit, survey, two land-ONs with releases, egress and landing 0.7 m from the launch point — was flown in **4 min 12 s** on 27 August.

3.3 Avionics and sensor subsystem

Subsystem	Component	Function and status
Flight controller	Holybro Pixhawk 6X, PX4 v1.17.0	SYS_AUTOSTART = 6001 (generic hexarotor X), CA_ROTOR_COUNT = 6, PWM_MAIN_FUNC1..6 = 101..106 — verified pin by pin with ACTUATOR_TEST, all six motors observed spinning in the correct order and direction
Companion computer	Raspberry Pi CM4	Mission orchestrator, vision, watchdog, GCS server; own power rail
Nadir camera	Meige OV9281 USB UVC, 1280 × 720 mono global shutter	Sole control-authority sensor: ArUco decode + white-pad cue. FOV measured 74.2° (fx ≈ 847 px). Hard-mounted, no gimbal.
Camera orientation	Mount yaw measured 180°	The camera is bolted upside down ; image row 0 looks at the tail. Measured from four independent placements of a floor target around the levelled airframe (180.3 / 187.4 / 183.4 / 186.1°, snapped to the bolted 180)
Height (primary)	Barometer , EKF2_HGT_REF = 0	GPS height reference was measured diverging 10.8 m peak-to-peak inside one flight and returned an aircraft that was tracking transit correctly — the barometer is the reference at both fields
Height (terminal)	Benewake TFmini-S downward lidar, TELEM3 (SENS_TFMINI_CFG = 103)	EKF2_RNG_CTRL = 1, conditional aiding below EKF2_RNG_A_HMAX = 7.0 m — pins the final metres of the descent
Optical flow	None (EKF2_OF_CTRL = 0)	No flow module in the kit; the decision is pinned so it cannot silently re-enable
GNSS	Non-RTK GPS/compass	Measured 14–16 satellites, horizontal accuracy ≤ 2 m at the practice field
Power sensing	Holybro PM02D , avionics only	BAT1_CAPACITY = -1 forces the voltage-only gauge; BAT1_V_DIV closed by multimeter (24.9 V at the connector vs 24.89 V reported — 0.04 %)
ESC telemetry	None — physically absent	The ESCs are PWM-only with no telemetry lead; DSHOT_TEL_CFG = 0 and zero ESC_STATUS even after the FC accepted a message-interval request. Per-motor current is the only signal a motor-out detector could use, so both detector layers were deleted rather than left inert ; the safety pilot is the mitigation, and restoring the feature starts with buying telemetry-capable ESCs
Imaging downlink	FPV camera + 5.8 GHz VTX	The “transmit” half of the imaging rule, independent of WiFi

Why the camera pose is load-bearing. With no gimbal the camera pitches with the body, so vision/projection.py composes roll and pitch into every pixel→ground fix; a translating multirotor holds 10–15° of tilt, which uncompensated

is ~2 m of error at 12 m altitude. Both the mount yaw and the lens FOV had shipped as **assumptions** and both were wrong — the yaw by ~180°, which point-mirrored every fix about the aircraft and is exactly the signature of a sweep that sees pads and never re-acquires them. They are now bench measurements, and the simulator carries the same 180° mount so the same error would fail in simulation instead of at the field.

Exposure — the fix that made in-flight decoding work. The camera’s own auto-exposure meters the whole frame, which at this field is sunlit grass at ~130 counts; a white pad is four to five times that albedo and clips at 255, and the marker’s black modules bleed into 1–3 px lines. That is why 2,788 frames of one flight decoded nothing with a pad plainly in view. The grabber now runs a **highlight-priority auto-exposure**: it meters the brightest 0.2 % of the frame every 0.5 s and steps the manual exposure to hold that between **190 and 225**, caps the ground mean at 55 when nothing bright is in view, and never exceeds a 4 ms exposure (about 1.3 px of smear at 3 m/s from 8 m). A marker carried at running speed across 8–10 m decoded **56–65 % of frames** at 25–35 px on the bench — the sweep’s own pixel size at twice its angular rate — and the 26 August flight decoded ids in the air for the first time.

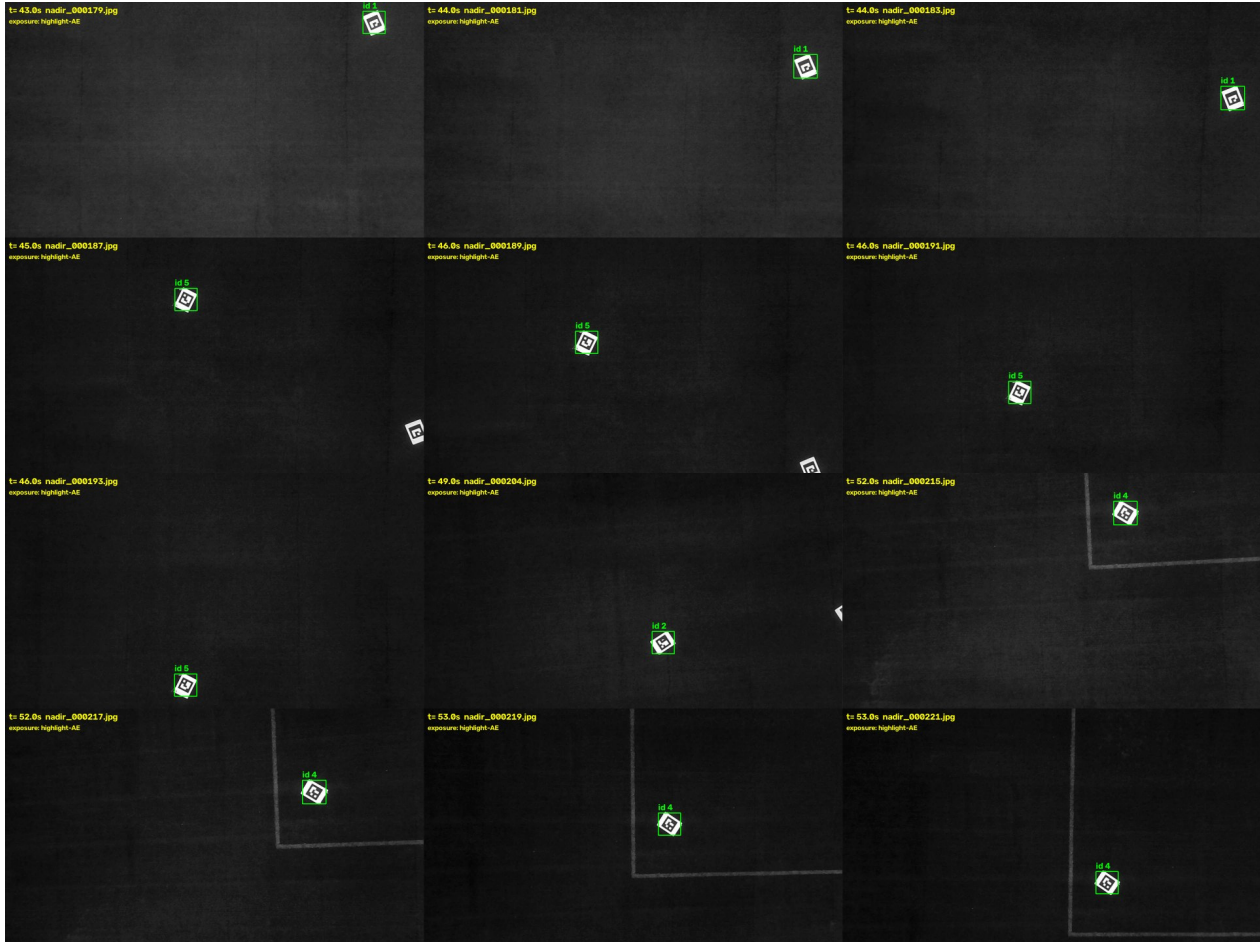


Figure 4. Real flight, 26 August 2026: twelve nadir frames in which the detector decoded a pad **in the air**, with the marker outlined and the id it read. Markers 1, 5, 2 and 4 appear here. Every frame is labelled exposure: highlight-AE — this is the flight immediately after the exposure fix, on the same camera and the same field that had produced 2,788 frames and zero decodes two flights earlier.

3.4 Software system architecture — operating modes and safety features

The autonomy stack is a **deterministic, offline, classical** system on the CM4 (Python 3.12, async throughout, MAVSDK to the flight controller). There is no LLM, no cloud and no network in flight — network use is an automatic disqualification. Layers:

- **Mission orchestrator** — the state machine of §2.3; one long-running process flies every flight of the window, holding in PREFLIGHT before each.
- **Live plan builder** — rebuilds the plan per flight and re-renders it at every gate release and every serve, so the ground station always shows what the aircraft is actually going to do next.
- **Vision worker** — ArUco decode plus the white-pad blob cue, with a frame-age gate and multi-pad fixes per frame.
- **Pad registry** — id-keyed clusters with k-vote confirmation, median-fused geolocation, and a maximum fix dis-

tance so a wild projection cannot create a target.

- **Terminal controller** — the rung descent, the id-verified LAND gate, the centred-final-rung gate and the touchdown-gated release.
- **Safety watchdog** — a 2 Hz phase-aware monitor (below).
- **Time and energy policies** — the window reserve, the per-flight and per-delivery gates, and the pack budget with swap detection.
- **Audit trail and verifier** — every flight writes a machine-readable `audit.jsonl` (1 Hz telemetry samples, transit pass/miss per point per direction, flight and delivery start/release/end lines); a **fail-closed** post-flight verifier re-scores the flight against it.
- **Ground station** — a Svelte + FastAPI web GCS: live map with the corridor drawn, camera view, the live plan, the ordered 4-of-6 mission-queue editor, the per-flight GO, an emergency KILL, and a confirmed-pad readout that shows **each pad id as the ArUco glyph itself** so the operator matches the committee's card picture-to-picture instead of translating it to a number.

Operating modes: PREFLIGHT · TAKEOFF · TRANSIT_INGRESS · SEARCH · LOCALIZE · LAND · DROP · TRANSIT_EGRESS · RTH · ABORT.

Companion-side watchdog (2 Hz, phase-aware). Every check below is active in flight; none can be switched off:

Condition	Response
Geofence breach	Return to home (proximity inside the margin = warning)
No-fly-zone entry	Return to home
Altitude ceiling	Warning above 30.5 m ; sustained above 31.5 m → return to home
Below the 10 m search floor outside the delivery descent	Advisory anomaly
Terrain proximity / unknown terrain (lidar)	Advisory, escalating to return to home
Battery below 30 % (planned)	Stop starting anything new; fly the corridor egress home, land, disarm
Battery below 15 % / 10 %	Return to home / land in place
Battery telemetry NaN, sustained	Loud operator anomaly — never a silent loss of battery protection
GPS 3D-fix loss, debounced 5 s	Return to home
Datalink loss, debounced 5 s	Return to home
Telemetry stale > 10 s	Return to home
Camera frame older than 2 s	Treated as no detection — a frozen image can never satisfy the LAND gate
Time budget exhausted	Return to home
Pilot takeover (manual mode from the RC)	Orchestrator stands down ; no companion command may fight the pilot
FC-commanded RTL or LAND we did not ask for	Orchestrator stands down (FC FAILSAFE) on the first tick

The last two are recent and were both bought with real flights. On 26 August PX4's own vertical fence returned an aircraft that was flying correctly, and the mission never noticed: it kept sweeping, and its next waypoint went out while the RTL was 0.46 m from touchdown — PX4 obeyed and climbed back out. The watchdog now records which AUTO mode *we* commanded and treats any other FC-initiated RTL or LAND as a stand-down. On 27 August the first version of that rule fired on **our own** landing 0.3 s after the first real egg was released, so two further rules were added: a LAND while on the ground is never a failsafe, and an expectation is **consumed** when the flight controller leaves the mode by itself, never cleared ahead of time by the next command.

Flight-controller-level failsafes (these fire even if the companion dies; all are written at mission start and read back):

- NAV_DLL_ACT = Return (datalink loss) · NAV_RCL_ACT = Return (RC loss, with COM_RCL_EXCEPT so an autonomous no-RC flight is not spuriously returned) · GF_ACTION = **3, Return**.
- Battery: BAT_LOW_THR 25 % · BAT_CRIT_THR 15 % · BAT_EMERGEN_THR 7 % with COM_LOW_BAT_ACT = 3, arranged so the FC's critical threshold coincides with the companion's return-home floor and its emergency threshold sits under the companion's land floor — the two layers never race.
- RTL_RETURN_ALT = **25 m**: any failsafe RTL is a straight line home over the building, the display aircraft and the tree line, and 25 m clears them while leaving 5 m under the ceiling for PX4's in-flight home-altitude drift. PX4's 60 m default would bust the ceiling.
- GF_MAX_VER_DIST = 50 m as a gross-runaway net only: the FC's vertical fence is home-relative, and PX4 1.17 rewrites the home altitude in flight, so it cannot be trusted to hold the rules' ceiling. The ceiling is enforced by the companion against a latched home altitude.

Three defects found in flight and fixed — worth stating because they are the reason the numbers in this report can be trusted:

1. **The mission clock ran at 1/20 speed on the real aircraft** (26 August). Mission time is read from the vehicle's own clock, and a re-fed identical timestamp was resetting the wall-clock anchor, so every deadline — the 20-minute window, the delivery gates, the leg-progress guard, the touchdown timeouts — stretched twentyfold and none of them could have fired at KMITL. Flight time is now accumulated per epoch as `min(vehicle, wall)` and a re-fed sample moves nothing.
2. **PX4 rewrites `home.alt` in flight** (upstream issue in 1.17), which moves the reported relative altitude while the aircraft physically holds station: measured shifts of +3.29 / -4.65 / +0.91 / +1.17 / -3.18 m across five flights, and one of them returned a correctly-flying aircraft through the ceiling watchdog. The height used by the mission is now altitude - home altitude latched at arming, and home updates are refused once airborne.
3. **The geofence action had been “Hold”, not “Return”, for its whole life** — the setter wrote the wrong enum value while its own name, comment and error text said RTL. The breach response was therefore “stop and loiter outside the fence”. The lesson kept in the code: a read-back gate proves a value was stored, never that the value means what the caller thinks.

Software integrity evidence. 771 automated tests import the real modules — the mission loop, the terminal controller, the watchdog, the detector, the projection maths, the flight clock, the home latch — and all pass, with linting and static type-checking clean. Behavioural changes are test-driven; the SITL gate exercises the whole sequence end to end; and every real flight is re-scored afterwards by the fail-closed verifier.

3.5 Payload handling mechanism

Four independent latches carry four boxed eggs. Each is a metal-gear servo on a Pixhawk **AUX** output configured as a peripheral actuator (`PWM_AUX_FUNC n = 300 + n`), commanded with **MAV_CMD_DO_SET_ACTUATOR (187)** — PX4 has **no handler for `DO_SET_SERVO`**, so that command is never used. **PWM 1900 μ s = release, 1100 μ s = hold**; one latch needs `PWM_AUX_MAX = 2100` to reach its release angle.

payload_id (release order)	Rack position	AUX pin / actuator set
0	Front-left	4
1	Rear-right	1
2	Front-right	2
3	Rear-left	3

The wiring was verified **pin by pin on the aircraft** with `ACTUATOR_TEST` while disarmed, watching which corner moved — the paper table written before that test had the order wrong. The same session found the four AUX outputs sitting on **RC passthrough** functions, i.e. two egg latches wired to the roll and yaw sticks, which would have released eggs whenever the aircraft banked.

Safety properties of the release path:

- **Never airborne.** The release runs only after the flight controller reports a landing, plus a 2 s settle. If telemetry reads airborne the release is skipped and audited.
- **Never twice.** Releases are idempotent against a ledger keyed by the **mission-global delivery index**, so two deliveries inside one flight can never collide and a retry cannot double-fire.
- **Never the wrong hold on a recovery flight.** The crew does not re-rack eggs at a battery swap, so a recovery flight continues the wiring-order progression through the latches that **still hold an egg**, rather than restarting at slot 0 and firing two empty holds.
- **Manual release is gated too.** The ground station's manual DROP is refused while telemetry reads airborne unless the operator explicitly forces it.

Per DCOS the payload is securely retained in flight and the mechanism has no unsafe state: a latch failure leaves an egg aboard, which costs a delivery and nothing else. The rules' prohibitions on simultaneous multi-cargo release and on winching are both respected — eggs are released one at a time, on the ground, one per landing.

3.6 Communication system

- **RC and telemetry on one link:** a **RadioMaster NOMAD** dual-band system carries both the safety pilot's ELRS control link and the MAVLink telemetry the ground station reads (receiver: DBR4 diversity). This replaced a telemetry radio that failed at the field, and it means the GCS panels are fed by the radio rather than by WiFi — a WiFi dropout never freezes the display.
- **Imagery:** an FPV camera on its own 5.8 GHz VTX satisfies the “transmit” half of the rules' imaging requirement, and the on-board frame recorder satisfies “record” by writing a JPEG trail to the CM4's own disk. Neither

half depends on the WiFi link, so the live dashboard image is a debug convenience that can be switched off on competition day at no cost.

- **Frequencies:** ELRS/NOMAD and the VTX operate inside the recommended 920–925 / 2400–2500 / 5725–5850 MHz bands. **No 4G/LTE, no SIM, no internet** — their use is a disqualification, and the field network is the CM4’s own local access point.
- **On-board routing:** a MAVLink router fans the flight link to the orchestrator, the ground station, and an optional QGroundControl endpoint that **defaults to loopback**, so the armed aircraft’s control plane is never exposed unauthenticated on a field network.
- **Standby-area discipline (procedural).** The rules forbid activating radio control and telemetry while in the standby area, and this aircraft raises its own access point at boot. The procedure is therefore: **do not connect the battery until the launch point**, and keep the ground console closed while waiting.

3.7 Operating procedures (normal and emergency)

Before the field day. A parameter tool reads back the flight controller’s own state and splits it into a **STOP block** — the values that live on the board and must be correct before flying (airframe, motor map, battery gauge endpoints, height reference, sensor ports, fence) — and an informational block of values the orchestrator pins at mission start. This exists because the motor map was once found reading zero, i.e. the aircraft was unflyable and looked normal; the cause was never identified, so it is treated as a regression that can recur and is re-read every field day rather than trusted.

Normal operation. The operator enters the committee’s four assigned ids once, as an ordered queue, using the ArUco glyphs shown on screen. Then, per flight: the readiness board goes green (link, arm-ready, EKF, home, sensors, GPS fix, battery $\geq 25\%$ resting, on-ground, geofence, fresh camera frame) → the operator confirms the eggs are racked and the crew is clear → **one GO click** → the aircraft flies the whole flight autonomously → it lands and disarms at L&R. The window clock starts at the first GO; the gates manage the schedule and refuse a launch that cannot finish, unless the operator explicitly accepts the overtime penalty.

Battery swap and recovery flight. If the planned 30 % egress fires, the aircraft comes home through the scored corridor and disarms. The crew swaps to the second charged pack; the between-flights gate waits as long as the swap takes. The next GO launches a **recovery flight** which flies direct to the pads still owed — they are already in the registry — and fires only the latches that still hold eggs.

Emergency operation. The safety pilot can take manual control at any instant, and the orchestrator stands down the moment it detects the takeover. The ground station’s KILL cuts motors, guarded by an armed command session so that an accidental click cannot fire it. Automatic failsafes — the companion watchdog and the FC-level RTL on geofence, datalink, RC and battery — recover the aircraft without operator action. On any unhandled mission error the orchestrator commands an emergency return and land, and the FC failsafe sits underneath that. Retries within the window are free under the rules, and the mission’s own registry makes a restart cheap: the pads are already known.

Brief for the safety pilot. A commanded mode change may need to be repeated once: the orchestrator stands down on the first tick, but a command already in the MAVLink pipeline (~ 0.5 s) can still reach the aircraft.

3.8 Airworthiness — DCOS compliance summary (rules Appendix B)

DCOS criterion	Compliance
Flight-performance margin across all load configurations	T/W ≈ 3.2 : 1, hover at $\sim 31\%$ thrust, loaded and empty (§3.2)
Envelope protection (tilt, rates, speed)	PX4 limits: 30° tilt, $25^\circ/\text{s}$ auto yaw, 5 m/s max horizontal, 1.5 m/s climb (§3.2)
Propulsion excess-thrust margin	$\sim 69\%$ throttle headroom at hover (§3.2)
Power reserve (no energy starvation)	Planned 4,700 mAh of 12,750 usable; planned 30 % corridor egress; FC battery failsafes below it; second pack + recovery flight (§3.2, §3.7)
Airframe load (flight / ground / transport)	§3.1 — envelope-limited manoeuvre load, 0.3 m/s touchdown, folding arms
Redundancy	Six rotors (rotor-out keeps attitude and thrust); two battery packs ; three height sources (lidar, barometer, GPS); FC and companion are independent computers, and the FC’s failsafes fire without the companion
Hardware compatibility (protocols, power, comms)	MAVLink throughout; separate avionics and motor power paths; §3.6 band compliance
Secure fastening / payload retention	Four independent latches, hold at 1100 μs , retained on failure (§3.5)
Clearance between moving parts	~ 43 mm prop-tip gap, no rotor overlap (§3.1)
Secure wiring and environmental protection	Strain-relieved looms; separate avionics rail; the companion on its own supply (§3.1)

DCOS criterion	Compliance
Envelope protection and fail-safe modes activatable at any time	Companion watchdog (2 Hz, non-optional) + FC failsafes + pilot takeover + GCS KILL (§3.4)
Normal and emergency procedures	§3.7
PPE per role	§1

Appendix A — Bill of materials, as built

The aircraft is not the one this report's version 1.1 costed. The frame, ESCs, propellers and both battery packs were **lent or donated by DRON (Defence Research Operation Network)**; the flight controller, companion computer, radio chain and ground equipment were already owned or were paid for by the team members themselves. The list below is what actually flies.

Group	Item	Part	Qty	Source
Airframe	Frame	EFT X6100 hexa-X, 1.000 m wheelbase, folding carbon arms	1	DRON
Propulsion	Motors	EFT E5 5008, 335 KV	6	DRON
Propulsion	Propellers	18 × 6.5 in	6 (+ spares)	DRON
Propulsion	ESCs	6 × PWM ESC (no telemetry lead)	6	DRON
Power	Mission pack	6S 17,000 mAh semi-solid , 10 C, 1.5 kg	1	DRON
Power	Spare pack (swap / recovery flight)	6S 15,000 mAh	1	DRON
Power	Power module	Holybro PM02D — flight controller and avionics only	1	owned
Power	Companion supply	Dedicated power-bank rail for the CM4	1	owned
Avionics	Flight controller	Holybro Pixhawk 6X , PX4 v1.17.0	1	owned
Avionics	Companion computer	Raspberry Pi CM4 + carrier	1	owned
Avionics	GNSS / compass	Non-RTK GPS module (cable replaced after an intermittent fault)	1	owned
Sensing	Nadir camera	Meige OV9281 USB UVC, 1280 × 720 mono global shutter, 74.2° measured	1	owned
Sensing	Rangefinder	Benewake TFmini-S , TELEM3	1	owned
Imaging	FPV camera + 5.8 GHz VTX	“transmit” half of the imaging requirement	1	owned
Radio	RC + telemetry	RadioMaster NOMAD (dual-band) + DBR4 receiver + TX16S	1 set	owned
Payload	Egg latches	4 × metal-gear servo on AUX 4/1/2/3 + rack	4 (+1 spare)	team-built
Ground	GCS	Laptop running the team's web GCS; CM4 local access point	1	owned

Retired from the version-1.1 BOM and **not** carried: the X500 V2 quad frame, the 10-inch propulsion set, the camera gimbal servo, the ARK Flow optical-flow module and the TF-Luna rangefinder. Two of those are worth noting as deliberate subtractions rather than omissions — the gimbal (the camera is hard-mounted and the projection composes attitude instead) and optical flow (not in the kit, and pinned off so it cannot silently re-enable).

Appendix B — Validation evidence

B.1 Simulation (PX4 SITL + Gazebo, KMITL field geometry)

The full mission is flown in simulation against an independent, **fail-closed** post-flight verifier. Ground truth is used only for the audit — never for planning.

Run	Configuration	Result
22 Jul 2026	First hexacopter model, one egg per flight, 4 flights	4/4 delivered id-correct, releases 0.13–0.25 m , transit 8/8 in order, 14.8 min, verifier 19 checks / 0 warnings
25 Jul 2026	Four eggs in one flight , 6-pad field	4/4 delivered id-correct, releases 0.15–0.21 m , transit 6/6, max altitude 19.69 m, landed 2.6 m from L&R, 457 s , verifier 14 / 0
24 Aug 2026	Competition envelope (transit 20 m, ceiling 20, sweep 12 m — the 28 px decode case), 4 eggs	6/6 pads found, 6/6 ids correct, 4/4 eggs , releases 0.09–0.22 m , transit 6/6, landed 2.0 m from L&R, 536 s of 1200

The 24 August pair was flown head-to-head to settle the sweep overlap with numbers rather than argument: 0.30 and 0.44 overlap on the same pad layout gave **identical coverage and identical delivery success**, with 0.30 **111 s faster** there and a computed 158 s faster on the KMITL polygon. Decode was separately measured at every across-track offset out to 94 % of the half-swath, so the wider spacing never puts a pad where the detector fails.

The same session produced the most instructive failure in the project: with the simulator's camera still mounted at yaw 0 against a configuration that had just been set to the **measured** 180°, the aircraft **decoded all six pads with every id correct and delivered 0 of 4**, flying to empty grass 3.7–20.2 m from each pad. That is the signature of a point-mirrored projection, and it is now asserted by a test so the two mounts can never disagree again.

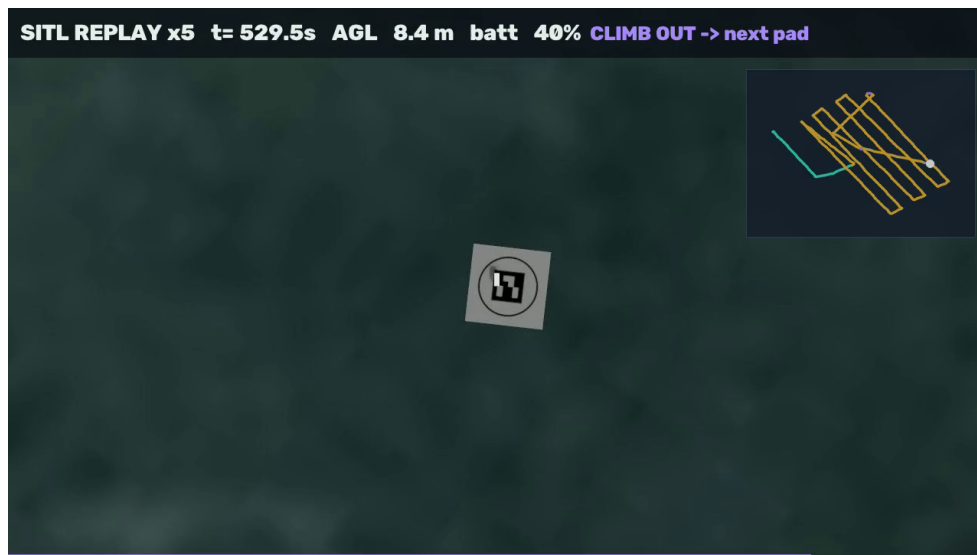


Figure 5. Simulation replay of a full four-egg mission. The main image is the aircraft's own nadir camera over a pad at 8.4 m; the overlay carries mission time, AGL, battery and phase, and the inset shows the flown track — the transit corridor in green and the completed survey sweep in orange. Ground truth from the simulator is used only for the post-flight audit, never for planning.

B.2 Real flights (KMUTNB practice field, standing in for KMITL)

Date / time	Duration	What was flown	Outcome
20 Aug	3 flights	First outdoor autonomy: transit + sweep	Found: every waypoint was sent with an undefined yaw, so PX4 turned the nose at each one — 867° of yaw in 122 s , and the body-fixed camera spun with it (1 of 457 frames decoded). Found: the GPS height reference inflated the reported AGL and returned a flight that was tracking transit to 1.7 m. Both fixed (single sweep heading; barometric height reference)
21 Aug	bench, props off	Pilot-takeover drill on the real flight controller and companion	Stand-down at the 1 s debounce verified; no companion command can fight the pilot
24 Aug	2 flights (from the FC's own log)	Take-off gate investigation	Found: PX4 leaves AUTO_TAKEOFF as soon as it is within NAV_MC_ALT_RAD of the target, levelling off 0.8 m low and aborting an otherwise perfect mission at take-off

Date / time	Duration	What was flown	Outcome
26 Aug	3 flights	Sweep with the new envelope	Found: PX4's own vertical fence returned a correctly-flying aircraft (its home altitude had walked 14 m in 93 s), the mission did not notice, and its next waypoint reached the aircraft while the return was 0.46 m from touchdown . Found: 2,788 frames, 0 decodes , with a pad plainly in view — the white pad was clipped at 255 by an exposure metered on grass. Both fixed (FC-failsafe stand-down, home-altitude latch, highlight-priority exposure); ids decoded in the air later the same day
27 Aug 12:44	5 min 56 s	Full 7-leg survey , 6,044 frames	5 of 6 pads decoded and confirmed (votes 13–14, confidence 0.95). The sixth pad is printed with marker id 0 and was being filtered out — the reason the valid set is now 0–6
27 Aug 14:13	—	Survey + serve	Confirmed all three of its assigned ids at t = 52 s and kept sweeping to leg 5 of 7. The sweep now stops as soon as every id the flight owes is confirmed
27 Aug 14:51	—	First delivery attempt	First real egg placed on a pad: 0.14 m from centre , released after touchdown. Exposed a false failsafe on our own landing, fixed the same day
27 Aug 15:25	4 min 12 s	Complete mission, end to end	Transit ingress 3/3 → survey found 5 pads in ~20 s → early stop on the assigned ids → land-ON pad 1, release; land-ON pad 4, release → egress → landed 0.7 m from the launch point → disarm. 2/2 delivered . Aircraft health: 14–16 satellites, horizontal accuracy ≤ 2 m, companion CPU 41 %, hover throttle 0.59–0.64, allocator never saturated, and the home-altitude latch held silent through five home rewrites

B.3 What the flight record establishes

- The **full sequence works on real hardware**: gate, transit, survey, registry, id-verified descent, land-ON, touchdown-gated release, egress, landing, disarm.
- **Release accuracy on the real aircraft (0.14 m)** is consistent with the simulated 0.09–0.30 m band, so the terminal controller is not simulator-flattered.
- Every safety layer added since 20 August has been **exercised in flight**, not only in tests: pilot takeover, FC-failsafe stand-down, the home-altitude latch, the early-stop sweep, and the release idempotence.
- The **remaining unknown is endurance**, not capability (§3.2, Appendix C).

Appendix C — Open items and risk register

Stated in the order they matter on competition day.

#	Item	Status and mitigation
1	Battery endurance vs the KMITL plan	The indicated gauge reaches the 15 % return floor at roughly minute 8.5–10; the plan needs 10–13 minutes. Mitigation, all implemented: per-delivery time-and-battery gate (gives up eggs, not the aircraft), planned 30 % egress through the scored corridor, second charged pack, automatic recovery flight from the unfired latches. Action: record the mAh returned by the charger after every flight — it is the one measurement that closes the question
2	The 28 Aug briefing changes are unflown	30 m ceiling, new corridor P2/P3', cut search area, 20 m survey with 10.5 m decode visits, 25 m failsafe RTL. The 13:00–18:00 trial slot is the only test before scoring
3	Committee coordinates not published	Our corridor and no-fly polygons are georeferenced from briefing photographs to ±5 m . Committee-published coordinates replace ours in both the flight configuration and the ground station, same digits in both
4	Where the pads are placed	The flown search area is cut at E 110 m, west of the main building. Pads placed east of the building would not be searched. Question to ask the committee before the scored flights; routing around the building box is a day-2 change
5	RC-loss drill	The parameters are pinned and read back, but the net itself has never been exercised in the air. The diagnosis of an earlier “no pulses” indication is closed (a held kill switch, not the link)

#	Item	Status and mitigation
6	CG record and structural margin	Not formally documented (§3.1). The aircraft is trimmed by flight behaviour; the cargo boxes are heavier than the dummies in the weighed figure, so it is re-weighed with real cargo at the field
7	Motor-map regression	The motor map was once found reading zero with no identified cause. It is re-read from the board before every field day rather than trusted
8	Marker id 0 in the field configuration	The flight detector accepts ids 0–6 (correct). The marker <code>.valid_ids</code> list in the field configuration still reads 1–6; it is consumed only by the simulator’s pad spawner, so it cannot affect a real flight, but it should be widened for consistency
9	Sim-only descent detail	The per-rung descent ladder writes the manual-mode descent parameter rather than its AUTO twin; the effective pad approach is the pinned 0.4 m/s that every validated landing has flown. Do not “unpin” it before the competition

Report structured to AAVC 2026 Rules & Regulations V1.3, Appendix A, as amended by the 24 July and 28 August 2026 event briefings. The flight system is deterministic, classical-CV and offline: no machine learning in the flight loop, no cloud, and no network — network use is an automatic disqualification. Supporting documents in this repository: docs/RULES_AAVC2026.md (rules digest), docs/FLIGHT.md (field procedure), docs/SERVO_AUX_MAPPING.md (payload wiring), docs/evidence/ (flight and bench evidence).